Cover Page

Data and Web Analytics Coursework
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23/07/2024

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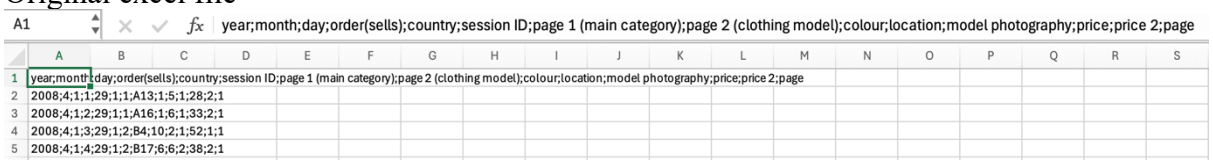
Task a:

Original excel file

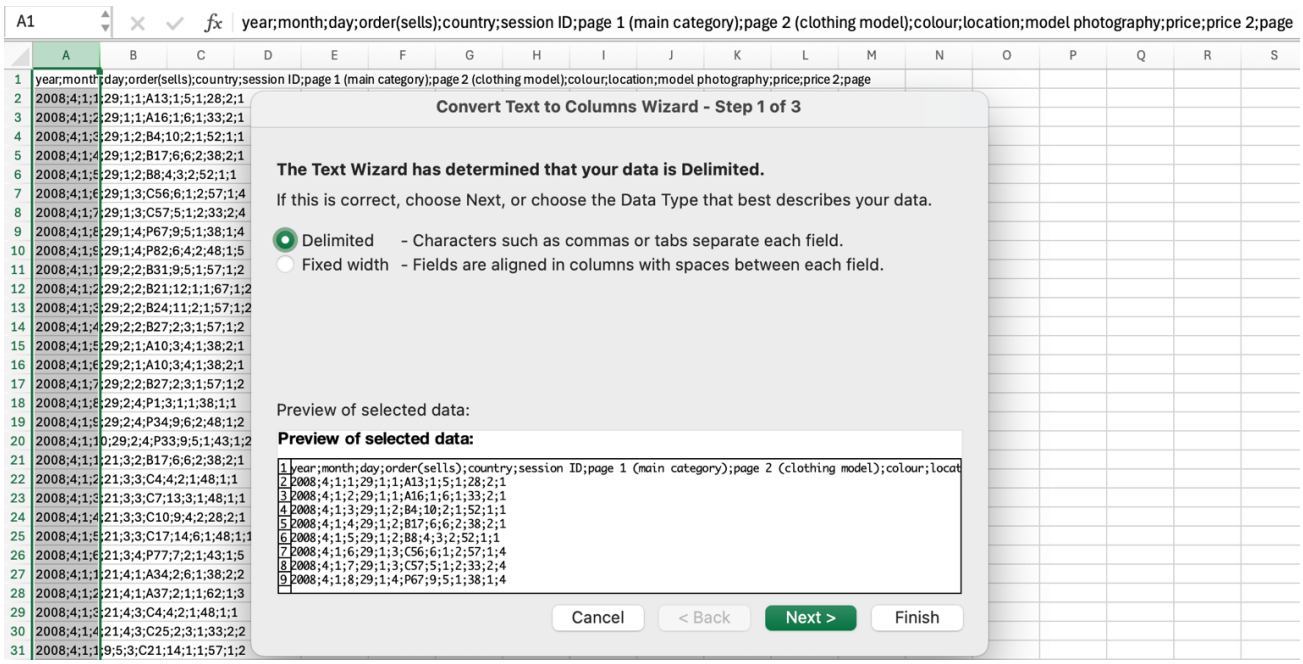


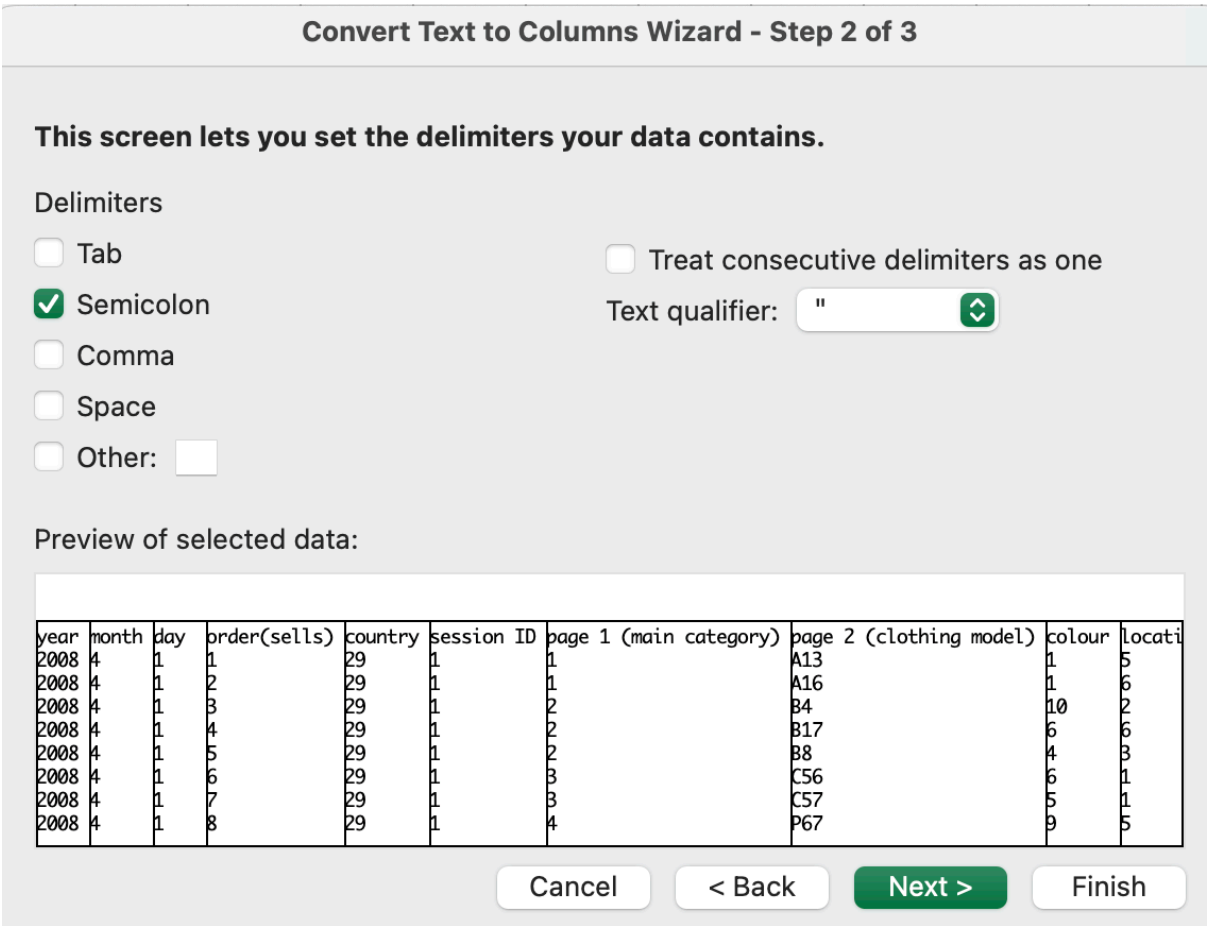
A1 | year;month;day;order(sells);country;session ID;page 1 (main category);page 2 (clothing model);colour;location;model photography;price;price 2;page

	A
1	year;month;day;order(sells);country;session ID;page 1 (main category);page 2 (clothing model);colour;location;model photography;price;price 2;page
2	2008;4;1;1;29;1;1;A13;1;5;1;28;2;1
3	2008;4;1;2;29;1;1;A16;1;6;1;33;2;1
4	2008;4;1;3;29;1;2;B4;10;2;1;52;1;1
5	2008;4;1;4;29;1;2;B17;6;6;2;38;2;1

2 ways to separate data:

1. in Excel





	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	year	month	day	order(sells)	country	session ID	page 1 (mai	page 2 (clot	colour	location	model pho	price	price 2	page
2	2008	4	1	1	29	1	1	A13	1	5	1	28	2	1
3	2008	4	1	2	29	1	1	A16	1	6	1	33	2	1
4	2008	4	1	3	29	1	2	B4	10	2	1	52	1	1
5	2008	4	1	4	29	1	2	B17	6	6	2	38	2	1
6	2008	4	1	5	29	1	2	B8	4	3	2	52	1	1
7	2008	4	1	6	29	1	3	C56	6	1	2	57	1	4
8	2008	4	1	7	29	1	3	C57	5	1	2	33	2	4
9	2008	4	1	8	29	1	4	P67	9	5	1	38	1	4
10	2008	4	1	9	29	1	4	P82	6	4	2	48	1	5
11	2008	4	1	1	29	2	2	B31	9	5	1	57	1	2
12	2008	4	1	2	29	2	2	B21	12	1	1	67	1	2
13	2008	4	1	3	29	2	2	B24	11	2	1	57	1	2
14	2008	4	1	4	29	2	2	B27	2	3	1	57	1	2
15	2008	4	1	5	29	2	1	A10	3	4	1	38	2	1
16	2008	4	1	6	29	2	1	A10	3	4	1	38	2	1
17	2008	4	1	7	29	2	2	B27	2	3	1	57	1	2
18	2008	4	1	8	29	2	4	P1	3	1	1	38	1	1
19	2008	4	1	9	29	2	4	P34	9	6	2	48	1	2
20	2008	4	1	10	29	2	4	P33	9	5	1	43	1	2
21	2008	4	1	1	21	3	2	B17	6	6	2	38	2	1
22	2008	4	1	2	21	3	3	C4	4	2	1	48	1	1
23	2008	4	1	3	21	3	3	C7	13	3	1	48	1	1
24	2008	4	1	4	21	3	3	C10	9	4	2	28	2	1
25	2008	4	1	5	21	3	3	C17	14	6	1	48	1	1
26	2008	4	1	6	21	3	4	P77	7	2	1	43	1	5
27	2008	4	1	1	21	4	1	A34	2	6	1	38	2	2

2. by coding in R

```r
{r}
install.packages("ggplot2")
install.packages("readr")
install.packages("readxl")

library(ggplot2)
library(readr)
library(readxl)
```

```r
{r}
# Read the Excel file
excel_content <- read_excel("e-shop clothing 2008.xlsx")


# Split the 1st row to create all column names
all_columns = strsplit(colnames(excel_content), ";")[[1]]

# Split the data by semicolon (;) delimiter
library(dplyr)
library(tidyr)
df <- excel_content %>%
  separate(col = colnames(excel_content)[1],
           into = c(all_columns),
           sep = ";", fill = 'right')
df
```

Columns were separated

A tibble: 165,474 × 14

year <chr>	month <chr>	day <chr>	order(sells) <chr>	country <chr>	session ID <chr>	page 1 (main category) <chr>	page 2 (clothing model) <chr>	colour <chr>	location <chr>	model photography <chr>	price <chr>	price 2 <chr>	page <chr>
2008	4	1	1	29	1	1	A13	1	5	1	28	2	1
2008	4	1	2	29	1	1	A16	1	6	1	33	2	1
2008	4	1	3	29	1	2	B4	10	2	1	52	1	1
2008	4	1	4	29	1	2	B17	6	6	2	38	2	1
2008	4	1	5	29	1	2	B8	4	3	2	52	1	1
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2008	4	1	7	29	1	3	C57	5	1	2	33	2	4
2008	4	1	8	29	1	4	P67	9	5	1	38	1	4
2008	4	1	9	29	1	4	P82	6	4	2	48	1	5
2008	4	1	1	29	2	2	B31	9	5	1	57	1	2
2008	4	1	2	29	2	2	B21	12	1	1	67	1	2
2008	4	1	3	29	2	2	B24	11	2	1	57	1	2
2008	4	1	4	29	2	2	B27	2	3	1	57	1	2
2008	4	1	5	29	2	1	A10	3	4	1	38	2	1
2008	4	1	6	29	2	1	A10	3	4	1	38	2	1
2008	4	1	7	29	2	2	B27	2	3	1	57	1	2
2008	4	1	8	29	2	4	P1	3	1	1	38	1	1
2008	4	1	9	29	2	4	P34	9	6	2	48	1	2
2008	4	1	10	29	2	4	P33	9	5	1	43	1	2
2008	4	1	1	21	3	2	B17	6	6	2	38	2	1
2008	4	1	2	21	3	3	C4	4	2	1	48	1	1
2008	4	1	3	21	3	3	C7	13	3	1	48	1	1
2008	4	1	4	21	3	3	C10	9	4	2	28	2	1
2008	4	1	5	21	3	3	C17	14	6	1	48	1	1
2008	4	1	6	21	3	4	P77	7	2	1	43	1	5
2008	4	1	1	21	4	1	A34	2	6	1	38	2	2
2008	4	1	2	21	4	1	A37	2	1	1	62	1	3
2008	4	1	3	21	4	3	C4	4	2	1	48	1	1
2008	4	1	4	21	4	3	C25	2	3	1	33	2	2
2008	4	1	1	9	5	3	C21	14	1	1	57	1	2
2008	4	1	1	9	6	3	C7	13	3	1	48	1	1
2008	4	1	2	9	6	3	C15	6	5	1	33	2	1
2008	4	1	3	9	6	3	C53	14	6	1	38	2	3
2008	4	1	4	9	6	2	B17	6	6	2	38	2	1
2008	4	1	5	9	6	2	B26	13	3	1	57	1	2
2008	4	1	1	29	7	1	A11	3	4	1	62	1	1
2008	4	1	2	29	7	3	C5	11	2	1	48	1	1
2008	4	1	3	29	7	4	P60	14	2	1	28	2	4
2008	4	1	4	29	7	4	P56	2	1	1	33	2	4
2008	4	1	5	29	7	4	P55	14	1	2	33	2	4

1–40 of 165,474 rows

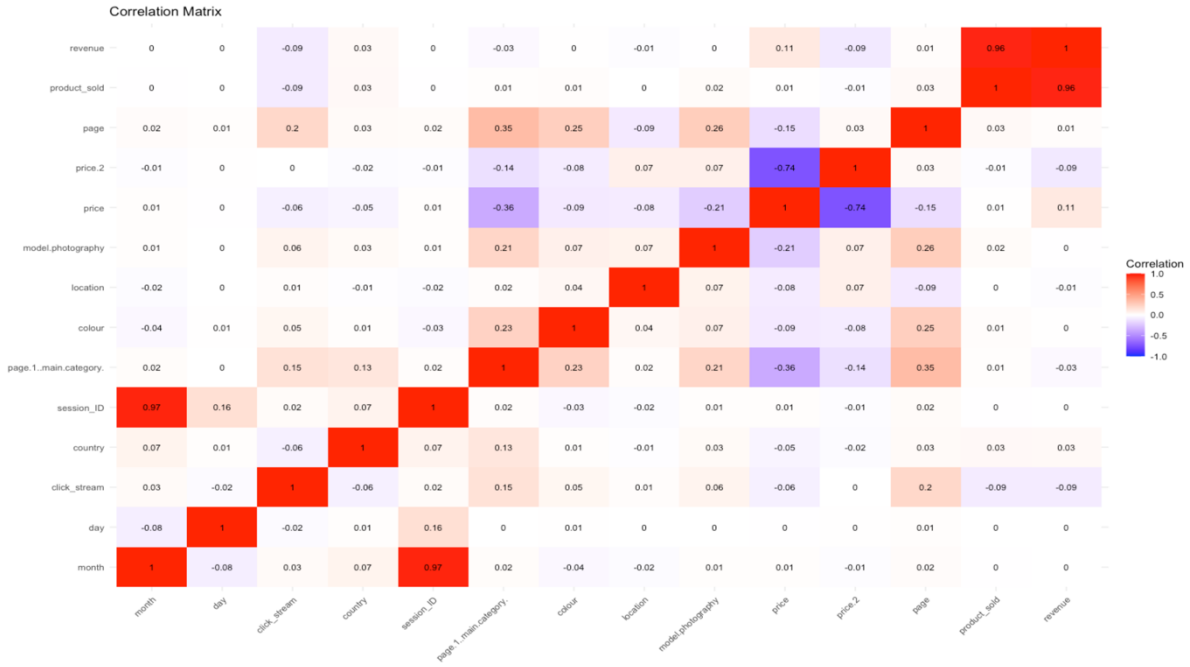
Previous 1 2 3 4 5 6 ... 25 Next

Extracted Columns:

1. **Month**: Helps identify seasonal trends and monthly variations in sales, revealing if sales peak during specific months, such as during summer sales or back-to-school shopping periods.

2. **Day**: The specific day of the month can be used to analyze daily trends, identify patterns related to weekdays vs. weekends, and pinpoint specific high or low sales days.
3. **Order(sells)/Click_stream**: Represents the sequence of clicks during one session, which is crucial for understanding customer behavior, click paths, and navigation patterns within the e-store.
4. **Country**: Identifies the country of origin of the IP address, providing insights into the geographical distribution of customers, which can influence marketing strategies, inventory distribution, and localization of the e-store.
5. **Session ID**: This unique identifier for each session allows tracking of individual user sessions, essential for session analysis, understanding user behavior, and measuring engagement levels.
6. **Page 1 (Main Category)**: Knowing the main product category that customers are interested in helps analyze product category performance, identify popular categories, and make inventory decisions.
7. **Colour**: Analyzing color preference helps understand customer preferences, trends, and aids in inventory planning for popular colors.
8. **Location**: The photo location on the page impacts customer engagement and product visibility, useful for optimizing website layout and product placement.
9. **Model Photography**: Indicates whether the product is shown in a front view or profile, which can affect customer perception and engagement. Analyzing this can help in optimizing product photography for better sales.
10. **Price**: Fundamental for revenue analysis, pricing strategy evaluation, and understanding the impact of price changes on sales.
11. **Price 2**: Indicates whether the price is higher than the average price for the product category, useful for pricing strategy analysis and understanding the market positioning of products.
12. **Page**: The page number within the e-store helps analyze navigation patterns, customer engagement across different pages, and optimize the e-store structure for better user experience.
13. **Revenue**: This is a key metric for understanding the financial performance of the e-store. It helps analyze the total sales generated, identify high-revenue products, and assess the overall profitability of the business.

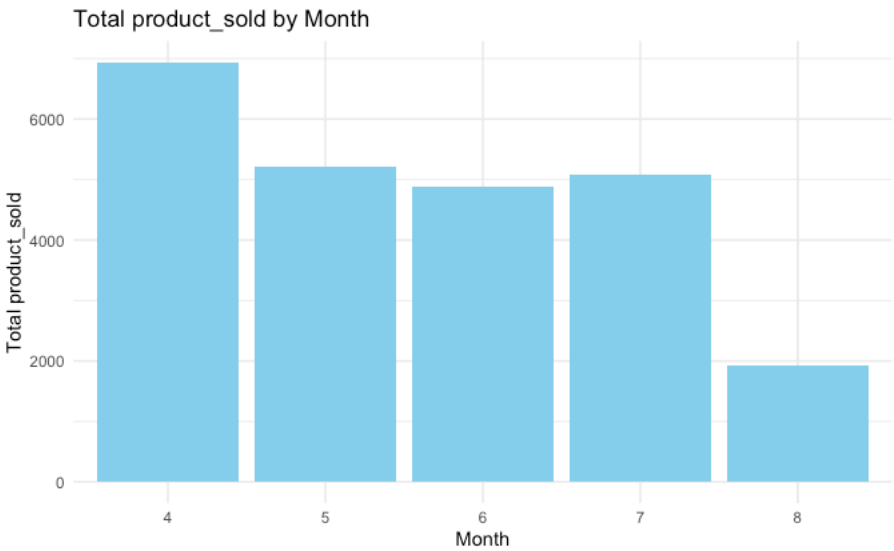
Correlation matrix



Explain

Task b:

Sales by Month Chart



The bar chart titled "Total product_sold by Month" shows the number of products sold during each month from April (4) to August (8). The chart reveals the following:

- **April (4)**:
  - Sales are approximately 6,900 units.
  - This is the highest sales month in the dataset, reflecting strong market demand or successful sales strategies.

- **May (5)**:
  - Sales decrease to roughly 5,200 units.
  - This is a decrease of 1,700 units from April, a 24.6% decline.
- **June (6)**:
  - Sales slightly decrease further to approximately 4,800 units.
  - This is a decrease of 400 units from May, a roughly 7.7% decline.
- **July (7)**:
  - Sales increased slightly to approximately 5,100 units.
  - This is an increase of 300 units from June, a roughly 6.25% increase.
- **August (8)**:
  - Sales drop significantly to approximately 1,900 units.
  - However, this drop is due to data being collected only until the 13th day of August.

```r
# Filter for only month "8" (August) and get unique days
august_days <- df %>%
  filter(month == 8) %>%
  select(day) %>%
  distinct() %>%
  arrange(day)

# Display the unique days in August
print(august_days)
```

A tibble: 13 × 1

day <int>
1
2
3
4
5
6
7
8
9
10
11
12
13

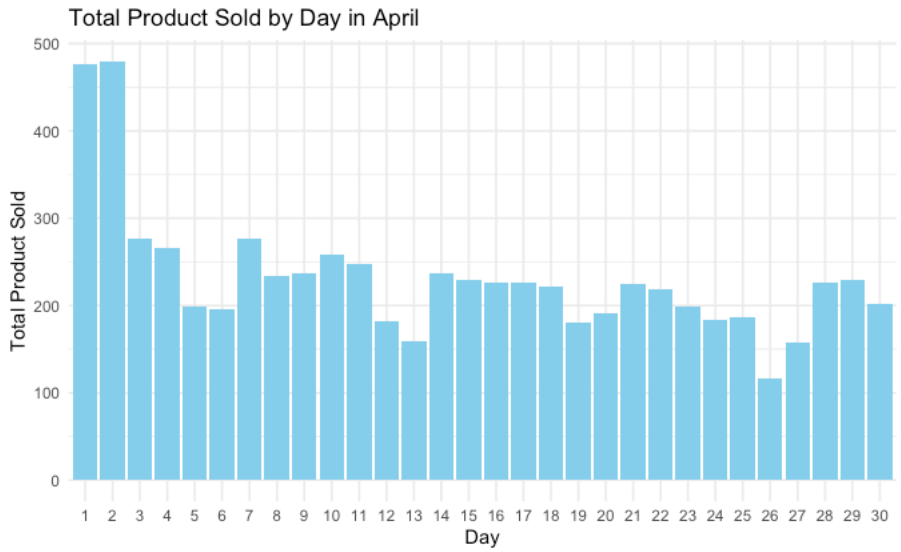
13 rows

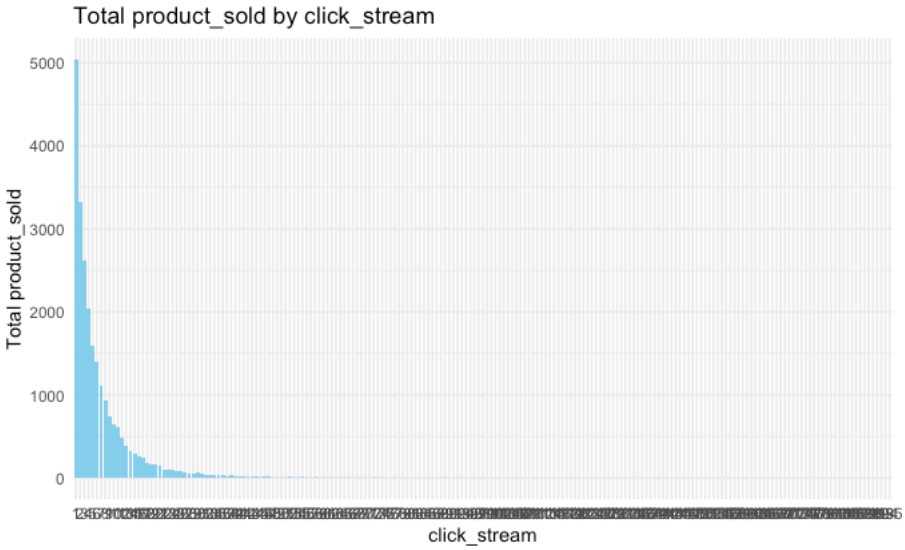
Summary for Sales Chart:

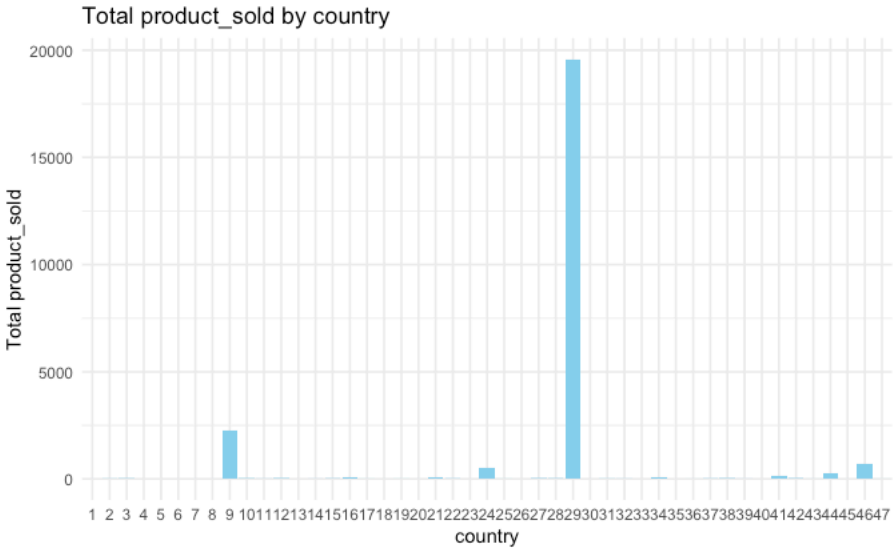
- The highest sales occur in April, followed by a sharp decline in May.
- Sales remain relatively stable from May to July.
- The significant drop in sales observed in August is primarily due to incomplete data, not necessarily a true reflection of market demand.

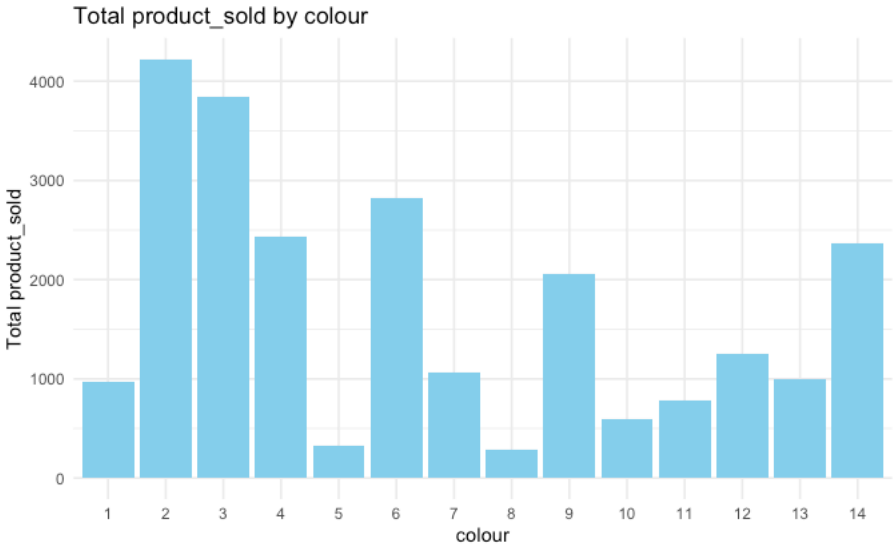
Possible Reasons:

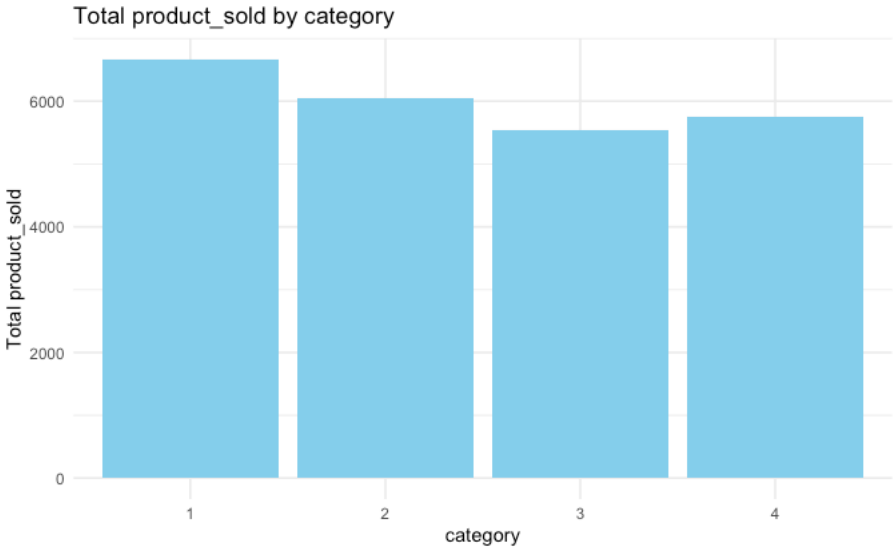
1. **Seasonal Demand**: April might have coincided with the launch of new products, a spring sale, or other promotional events, leading to a surge in sales. After April, the lack of such events or the end of a season could have led to the drop in sales. The relatively stable sales from May to July suggest consistent demand.
2. **Marketing and Promotions**: April's high sales could be a result of effective marketing campaigns or promotions that were not sustained in the following months.
3. **Inventory and Product Availability**: The decline after April might also suggest that popular products were out of stock or that the product lineup did not attract as much interest.
4. **Customer Behavior**: There could be a pattern in customer behavior where interest wanes as the summer progresses, possibly due to vacations or shifts in spending habits.
5. **Grand opening event**: The significantly high number of sold units on the first and second days in April, nearly double the sales of subsequent days, could be attributed to a grand opening event. This initial spike in sales is often seen during such promotions, as customers are drawn in by special offers and the excitement surrounding the launch.

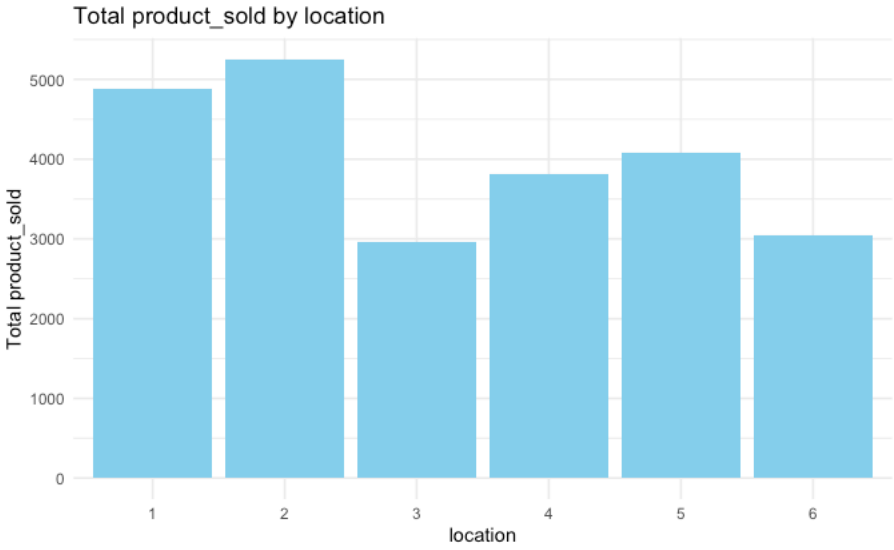


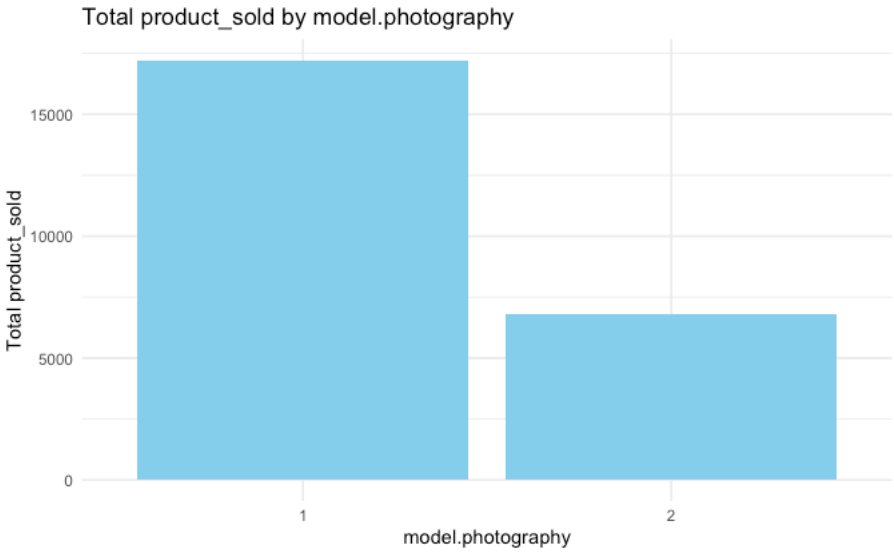

Total product_sold by click_stream
Total product_sold
5000
4000
3000
2000
1000
0
click_stream

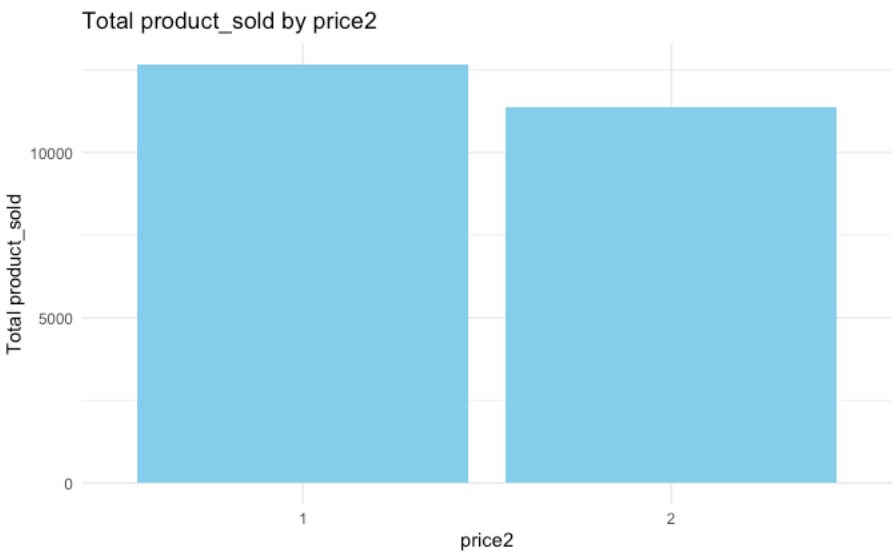

Total product_sold by country
Total product_sold
20000
15000
10000
5000
0
1 2 3 4 5 6 7 8 9 10111213141516171819202122232425262728293031323334353637383940414243444546 47
country

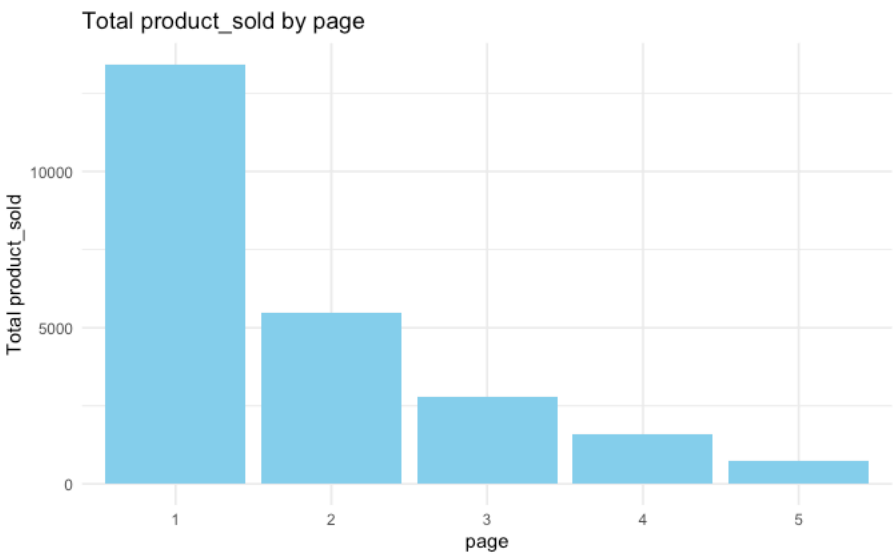

Total product_sold by colour
Total product_sold
4000
3000
2000
1000
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
colour


Total product_sold by category
Total product_sold
6000
4000
2000
0
1
2
3
4
category


Total product_sold by location
Total product_sold
5000
4000
3000
2000
1000
0
1
2
3
4
5
6
location


Total product_sold by model.photography
Total product_sold
15000
10000
5000
0
1
2
model.photography





Task c:

```
# Add a new column revenue
df <- df %>%
  mutate(revenue = product_sold * price)
df
```

A tibble: 165,474 × 14

month <int>	day <int>	click_stream <int>	country <int>	session_ID <int>	category <int>	colour <int>	location <int>	model.photography <int>	price <int>	price2 <int>	page <int>	product_sold <dbl>	revenue <dbl>
4	1	1	29	1	1	1	5	1	28	2	1	0	0
4	1	2	29	1	1	1	6	1	33	2	1	0	0
4	1	3	29	1	2	10	2	1	52	1	1	0	0
4	1	4	29	1	2	6	6	2	38	2	1	0	0
4	1	5	29	1	2	4	3	2	52	1	1	0	0
4	1	6	29	1	3	6	1	2	57	1	4	0	0
4	1	7	29	1	3	5	1	2	33	2	4	0	0
4	1	8	29	1	4	9	5	1	38	1	4	0	0
4	1	9	29	1	4	6	4	2	48	1	5	1	48
4	1	1	29	2	2	9	5	1	57	1	2	0	0
4	1	2	29	2	2	12	1	1	67	1	2	0	0
4	1	3	29	2	2	11	2	1	57	1	2	0	0
4	1	4	29	2	2	2	3	1	57	1	2	0	0
4	1	5	29	2	1	3	4	1	38	2	1	0	0
4	1	6	29	2	1	3	4	1	38	2	1	0	0
4	1	7	29	2	2	2	3	1	57	1	2	0	0
4	1	8	29	2	4	3	1	1	38	1	1	0	0
4	1	9	29	2	4	9	6	2	48	1	2	0	0
4	1	10	29	2	4	9	5	1	43	1	2	1	43
4	1	1	21	3	2	6	6	2	38	2	1	0	0
4	1	2	21	3	3	4	2	1	48	1	1	0	0
4	1	3	21	3	3	13	3	1	48	1	1	0	0
4	1	4	21	3	3	9	4	2	28	2	1	0	0
4	1	5	21	3	3	14	6	1	48	1	1	0	0
4	1	6	21	3	4	7	2	1	43	1	5	1	43
4	1	1	21	4	1	2	6	1	38	2	2	0	0
4	1	2	21	4	1	2	1	1	62	1	3	0	0
4	1	3	21	4	3	4	2	1	48	1	1	0	0
4	1	4	21	4	3	2	3	1	33	2	2	1	33
4	1	1	9	5	3	14	1	1	57	1	2	1	57
4	1	1	9	6	3	13	3	1	48	1	1	0	0
4	1	2	9	6	3	6	5	1	33	2	1	0	0
4	1	3	9	6	3	14	6	1	38	2	3	0	0
4	1	4	9	6	2	6	6	2	38	2	1	0	0
4	1	5	9	6	2	13	3	1	57	1	2	1	57
4	1	1	29	7	1	3	4	1	62	1	1	0	0
4	1	2	29	7	3	11	2	1	48	1	1	0	0
4	1	3	29	7	4	14	2	1	28	2	4	0	0
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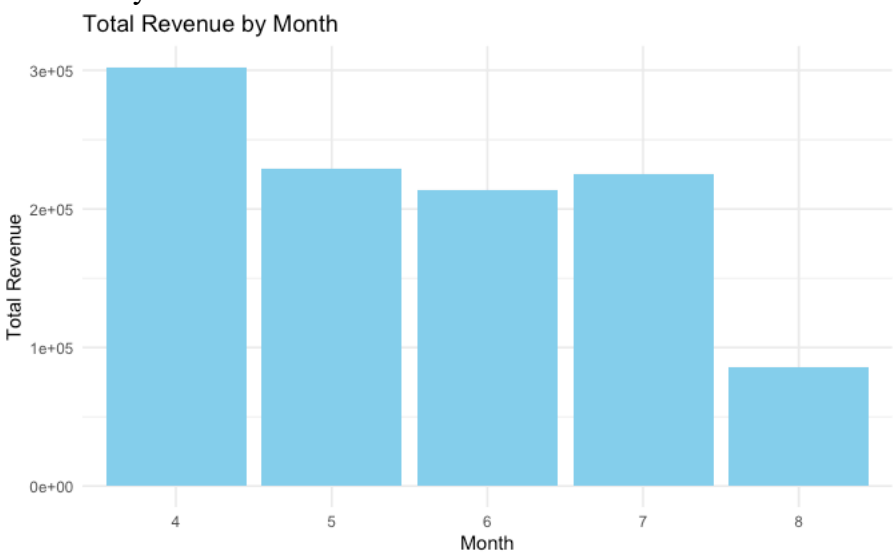
4	1	8	29	7	4	6	2	2	28	2	3	0	0
4	1	9	29	7	4	6	2	2	28	2	2	0	0
4	1	10	29	7	4	3	1	1	38	1	1	0	0

1–45 of 165,474 rows

Previous 1 2 3 4 5 6 ... 23 Next

Revenue by Month Chart



Analysis

- **April (4)**:
  - Revenue is approximately 300,000 US dollars.
  - This is the highest revenue month in the dataset, indicating strong performance.
- **May (5)**:
  - Revenue drops to approximately 220,000.
  - This is a decrease of 80,000 from April, a roughly 26.7% decline.
- **June (6)**:

    - Revenue continues to decline, reaching approximately 210,000.
    - This is a decrease of 10,000 from May, a roughly 4.5% decline.
- **July (7)**:
    - Revenue slightly increases to approximately 220,000.
    - This is an increase of 10,000 from June, a roughly 4.8% increase.
- **August (8)**:
    - Revenue drops significantly to approximately 180,000.
    - However, this drop is due to data being collected only until the 13th day of August.

Summary for Revenue Chart:

- The highest revenue occur in April, followed by a sharp decline in May.
- Revenue remain relatively stable from May to July.
- The significant drop in revenue observed in August is primarily due to incomplete data, not necessarily a true reflection of market demand.

Possible Reasons:

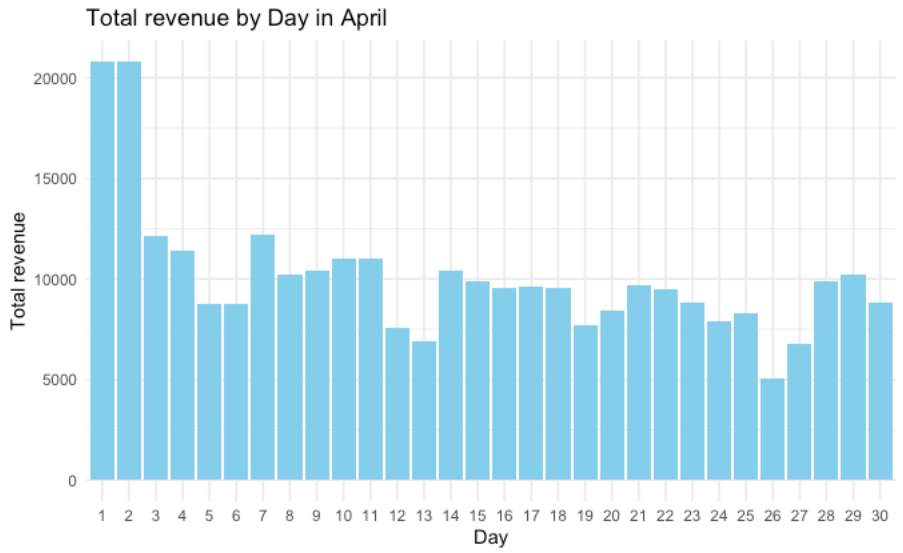
1. **Seasonal Demand**: April might have coincided with the launch of new products, a spring sale, or other promotional events, leading to a surge in revenue. After April, the lack of such events or the end of a season could have led to the drop in revenue. The relatively stable revenue from May to July suggest consistent demand.
2. **Marketing and Promotions**: April's high revenue could be a result of effective marketing campaigns or promotions that were not sustained in the following months.
3. **Inventory and Product Availability**: The decline after April might also suggest that popular products were out of stock or that the product lineup did not attract as much interest.
4. **Customer Behavior**: There could be a pattern in customer behavior where interest wanes as the summer progresses, possibly due to vacations or shifts in spending habits.
5. **Grand opening event**: The significantly high number of revenue on the first and second days in April, nearly double the revenue of subsequent days, could be attributed to a grand opening event. This initial spike in sales is often seen during such promotions, as customers are drawn in by special offers and the excitement surrounding the launch.

April

- **Strong Performance**: April had the highest revenue, possibly due to seasonal demand or successful marketing strategies that attracted more customers.

- **Promotional Campaigns**: Special promotions or events in April might have driven higher sales, contributing to the revenue peak.
- **Grand Opening Event:** The unusually high revenue recorded on the first and second days of April, nearly double that of the following days, may be linked to a grand opening event. Such promotional events typically result in an initial surge in sales, as customers are enticed by special deals and the excitement of the launch.

Revenue by Day in April Chart



May

- **Post-Peak Adjustment**: The drop in revenue in May (approximately 26.7%) may indicate a natural adjustment after the peak month of April, as customers may have exhausted their purchasing during the promotional period.
- **Increased Competition**: The entrance of competitors or alternative products may have drawn customers away, impacting sales.
- **Consumer Behavior**: Changes in consumer spending habits or seasonal trends may also contribute to the decline.

June

- **Continued Decline**: The slight decrease in June (approximately 4.5%) could suggest that the initial drop in May continued, potentially due to lingering effects of increased competition or reduced consumer interest.
- **Market Saturation**: By June, the market may have reached saturation, leading to less urgency in purchases from consumers.

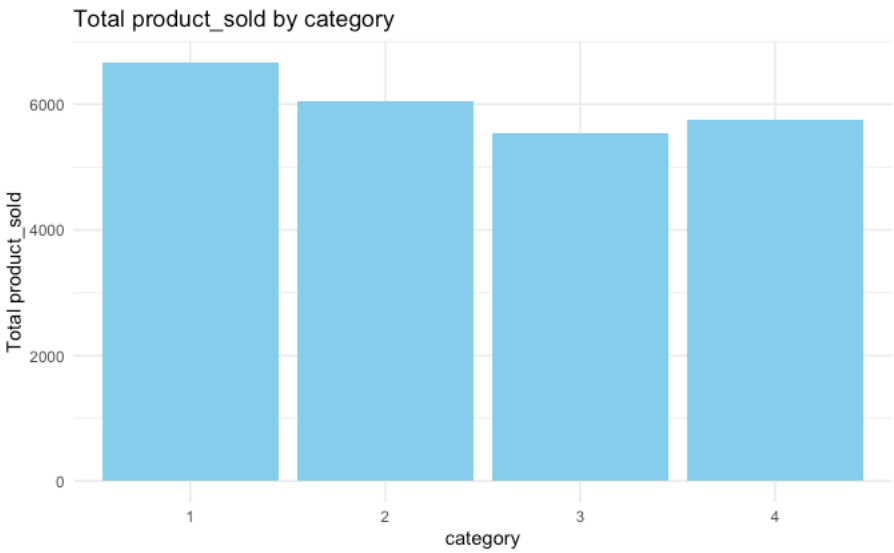
July

- **Minor Recovery**: The small increase in July (approximately 4.8%) could indicate a rebound as marketing efforts were adjusted or new products were introduced.
- **Seasonal Trends**: July may benefit from summer sales or promotions, attracting more customers and encouraging spending.

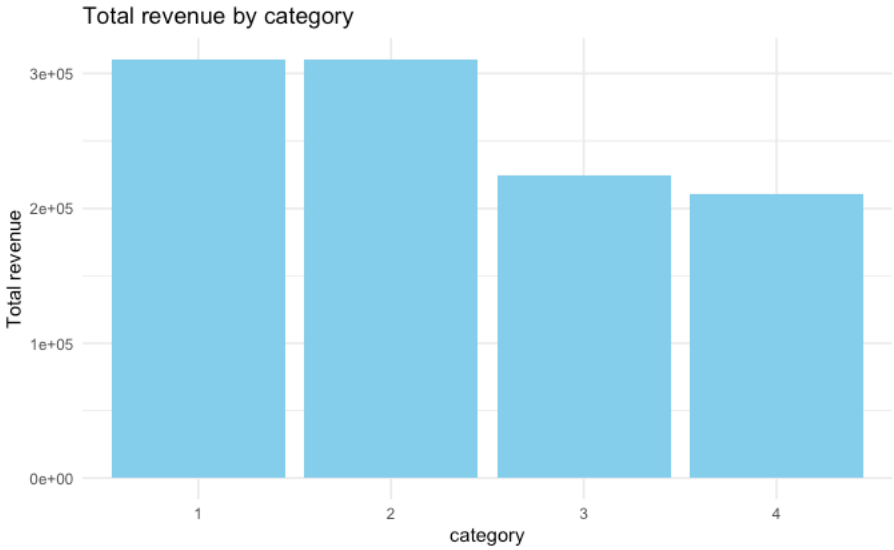
August

- **Limited Data**: The significant drop in August (to approximately 180,000) should be viewed with caution, as it only reflects data collected until the 13th day.
- **Seasonal Lull**: If data collection continued for the full month, revenue may have picked up again as seasonal promotions began or as consumers prepared for back-to-school shopping.

Total product sold by category chart



Total revenue by category chart



Insight on the 2 above charts:

Product Category	Recommendations

Trousers (Category 1)	Because many trousers were sold and brought good revenue so we just need to: - Maintain quality and availability to sustain this performance. - Introduce premium or specialized products to further boost revenue. - Cross-sell with other categories to increase sales in lower-performing categories.
Skirts (Category 2)	Because skirts were sold less than trousers but brought as high revenue as trousers so we should **increase selling skirts** by: - Promote through special collections or limited editions to capitalize on their strong revenue potential. - Expand the range with new styles to capture more market share. - Implement targeted marketing campaigns to increase visibility and sales.
Blouses (Category 3)	- Although many blouses were sold but its revenue is not as much as it should be so we should **increase price of blouses**
Sale Items (Category 4)	Because sale brought small profit so we should: **- reduce the number of items on sale**, - focusing on moving only slow-moving inventory to improve overall profitability. - Offer bundling strategies to clear inventory. - Run limited-time promotional events to drive urgency. - Review inventory to phase out outdated or unpopular items.

Task d:

Original excel file

A2 | fx | 1 1 Rick 623.30 2012-01-01 IT

	A	B	C	D	E	F	G	H
1	id	name	salary	start_date	dept			
2	1 1 Rick	623.30	2012-01-01	IT				
3	2 2 Dan	515.20	2013-09-23	Operations				
4	3 3 Michelle	611.00	2014-11-15	IT				
5	4 4 Ryan	729.00	2014-05-11	HR				
6	5 NA Gary	843.25	2015-03-27	Finance				
7	6 6 Nina	578.00	2013-05-21	IT				
8	7 7 Simon	632.80	2013-07-30	Operations				
9	8 8 Guru	722.50	2014-06-17	Finance				
10								

Data after being cleaned

	A	B	C	D	E
1	id	name	salary	start_date	dept
2	1	Rick	623,30	1/1/12	IT
3	2	Dan	515,20	23/9/13	Operations
4	3	Michelle	611,00	15/11/14	IT
5	4	Ryan	729,00	11/5/14	HR
6	5	Gary	843,25	27/3/15	Finance
7	6	Nina	578,00	21/5/13	IT
8	7	Simon	632,80	30/7/13	Operations
9	8	Guru	722,50	17/6/14	Finance
10					

Descriptive Statistics for Salary

```r
# Load the input dataset
file_path2 <- "input.xlsx"
data <- read_excel(file_path2)

# Check for non-numeric values and convert salary column to numeric
data$salary <- as.numeric(as.character(data$salary))

# Calculate descriptive statistics
mean_salary <- mean(data$salary, na.rm = TRUE)
median_salary <- median(data$salary, na.rm = TRUE)

# Function to calculate the mode
calculate_mode <- function(x) {
  freq <- table(x)
  max_freq <- max(freq)

  # Check if the maximum frequency is 1, indicating all values are unique
  if (max_freq == 1) {
    return(NaN)
  } else {
    return(as.numeric(names(freq)[freq == max_freq]))
  }
}

# Calculate the mode of the salary column
mode_salary <- calculate_mode(data$salary)

std_dev_salary <- sd(data$salary, na.rm = TRUE)
variance_salary <- var(data$salary, na.rm = TRUE)

# Print the results
cat("Descriptive Analysis:\n")
cat("Mean Salary:", mean_salary, "\n")
cat("Median Salary:", median_salary, "\n")
cat("Mode Salary:", mode_salary, "\n")
cat("Standard Deviation of Salary:", std_dev_salary, "\n")
cat("Variance of Salary:", variance_salary, "\n")
```

```
New names:
• `` -> `...6`
• `` -> `...7`
• `` -> `...8`
• `` -> `...9`
Descriptive Analysis:
Mean Salary: 656.8813
Median Salary: 628.05
Mode Salary: NaN
Standard Deviation of Salary: 103.0595
Variance of Salary: 10621.25
```

Summary:

- The mean salary is around 656.88, indicating the average salary among the employees.
- The median salary is 628.05, showing the middle point of the salary distribution.
- There is no single mode, as all salaries are unique.
- The standard deviation is approximately 103.06, indicating the dispersion of salaries around the mean.
- The variance is 10621.25, representing the spread of the salary data.

Task e:

i. Identify actors with eye colors other than black and a height exceeding 150 cm

```
{r}
# Load necessary libraries
library(dplyr)
library(ggplot2)

# Load the Star Wars dataset
data("starwars")

# i. Extract actors with specific eye color and height criteria
filtered_actors <- starwars %>% filter(eye_color != "black" & height > 150)

# Print the filtered actors
print(filtered_actors)
```

A tibble: 61 × 14

name <chr>	height <int>	mass <dbl>	hair_color <chr>	skin_color <chr>	eye_color <chr>	birth_year <dbl>	sex <chr>	gender <chr>	homeworld <chr>	species <chr>	films <list>	vehicles <list>
Luke Skywalker	172	77.0	blond	fair	blue	19.0	male	masculine	Tatooine	Human	<chr [5]>	<chr [2]>
C-3PO	167	75.0	NA	gold	yellow	112.0	none	masculine	Tatooine	Droid	<chr [6]>	<chr [0]>
Darth Vader	202	136.0	none	white	yellow	41.9	male	masculine	Tatooine	Human	<chr [4]>	<chr [0]>
Owen Lars	178	120.0	brown, grey	light	blue	52.0	male	masculine	Tatooine	Human	<chr [3]>	<chr [0]>
Beru Whitesun Lars	165	75.0	brown	light	blue	47.0	female	feminine	Tatooine	Human	<chr [3]>	<chr [0]>
Biggs Darklighter	183	84.0	black	light	brown	24.0	male	masculine	Tatooine	Human	<chr [1]>	<chr [0]>
Obi-Wan Kenobi	182	77.0	auburn, white	fair	blue-gray	57.0	male	masculine	Stewjon	Human	<chr [6]>	<chr [1]>
Anakin Skywalker	188	84.0	blond	fair	blue	41.9	male	masculine	Tatooine	Human	<chr [3]>	<chr [2]>
Wilhuff Tarkin	180	NA	auburn, grey	fair	blue	64.0	male	masculine	Eriadu	Human	<chr [2]>	<chr [0]>
Chewbacca	228	112.0	brown	unknown	blue	200.0	male	masculine	Kashyyyk	Wookiee	<chr [5]>	<chr [1]>
Han Solo	180	80.0	brown	fair	brown	29.0	male	masculine	Corellia	Human	<chr [4]>	<chr [0]>
Jabba Desilijic Tiure	175	1358.0	NA	green-tan, brown	orange	600.0	hermaphroditic	masculine	Nal Hutta	Hutt	<chr [3]>	<chr [0]>
Wedge Antilles	170	77.0	brown	fair	hazel	21.0	male	masculine	Corellia	Human	<chr [3]>	<chr [1]>
Jek Tono Porkins	180	110.0	brown	fair	blue	NA	NA	NA	Bestine IV	NA	<chr [1]>	<chr [0]>
Palpatine	170	75.0	grey	pale	yellow	82.0	male	masculine	Naboo	Human	<chr [5]>	<chr [0]>
Boba Fett	183	78.2	black	fair	brown	31.5	male	masculine	Kamino	Human	<chr [3]>	<chr [0]>
IG-88	200	140.0	none	metal	red	15.0	none	masculine	NA	Droid	<chr [1]>	<chr [0]>
Bossk	190	113.0	none	green	red	53.0	male	masculine	Trandosha	Trandoshan	<chr [1]>	<chr [0]>
Lando Calrissian	177	79.0	black	dark	brown	31.0	male	masculine	Socorro	Human	<chr [2]>	<chr [0]>
Lobot	175	79.0	none	light	blue	37.0	male	masculine	Bespin	Human	<chr [1]>	<chr [0]>
Ackbar	180	83.0	none	brown mottle	orange	41.0	male	masculine	Mon Cala	Mon Calamari	<chr [2]>	<chr [0]>
Qui-Gon Jinn	193	89.0	brown	fair	blue	92.0	male	masculine	NA	Human	<chr [1]>	<chr [1]>
Nute Gunray	191	90.0	none	mottled green	red	NA	male	masculine	Cato Neimoidia	Neimodian	<chr [3]>	<chr [0]>
Finis Valorum	170	NA	blond	fair	blue	91.0	male	masculine	Coruscant	Human	<chr [1]>	<chr [0]>
Padmé Amidala	185	45.0	brown	light	brown	46.0	female	feminine	Naboo	Human	<chr [3]>	<chr [0]>
Jar Jar Binks	196	66.0	none	orange	orange	52.0	male	masculine	Naboo	Gungan	<chr [2]>	<chr [0]>
Roos Tarpals	224	82.0	none	grey	orange	NA	male	masculine	Naboo	Gungan	<chr [1]>	<chr [0]>
Rugor Nass	206	NA	none	green	orange	NA	male	masculine	Naboo	Gungan	<chr [1]>	<chr [0]>
Ric Olié	183	NA	brown	fair	blue	NA	male	masculine	Naboo	Human	<chr [1]>	<chr [0]>
Quarsh Panaka	183	NA	black	dark	brown	62.0	male	masculine	Naboo	Human	<chr [1]>	<chr [0]>
Shmi Skywalker	163	NA	black	fair	brown	72.0	female	feminine	Tatooine	Human	<chr [2]>	<chr [0]>
Darth Maul	175	80.0	none	red	yellow	54.0	male	masculine	Dathomir	Zabrak	<chr [1]>	<chr [1]>
Bib Fortuna	180	NA	none	pale	pink	NA	male	masculine	Ryloth	Twi'lek	<chr [1]>	<chr [0]>
Ayla Secura	178	55.0	none	blue	hazel	48.0	female	feminine	Ryloth	Twi'lek	<chr [3]>	<chr [0]>
Ben Quadinaros	163	65.0	none	grey, green, yellow	orange	NA	male	masculine	Tund	Toong	<chr [1]>	<chr [0]>
Mace Windu	188	84.0	none	dark	brown	72.0	male	masculine	Haruun Kal	Human	<chr [3]>	<chr [0]>
Ki-Adi-Mundi	198	82.0	white	pale	yellow	92.0	male	masculine	Cerea	Cerean	<chr [3]>	<chr [0]>
Eeth Koth	171	NA	black	brown	brown	NA	male	masculine	Iridonia	Zabrak	<chr [2]>	<chr [0]>
Adi Gallia	184	50.0	none	dark	blue	NA	female	feminine	Coruscant	Tholothian	<chr [2]>	<chr [0]>
Saesee Tiin	188	NA	none	pale	orange	NA	male	masculine	Iktotch	Iktotchi	<chr [2]>	<chr [0]>
Yarael Poof	264	NA	none	white	yellow	NA	male	masculine	Quermia	Quermian	<chr [1]>	<chr [0]>
Mas Amedda	196	NA	none	blue	blue	NA	male	masculine	Champala	Chagrian	<chr [2]>	<chr [0]>
Gregar Typho	185	85.0	black	dark	brown	NA	NA	NA	Naboo	NA	<chr [1]>	<chr [0]>
Cordé	157	NA	brown	light	brown	NA	NA	NA	Naboo	NA	<chr [1]>	<chr [0]>
Cliegg Lars	183	NA	brown	fair	blue	82.0	male	masculine	Tatooine	Human	<chr [1]>	<chr [0]>

1–45 of 61 rows | 1–13 of 14 columns

Previous 1 2 Next

ii. The procedure for adding a Body Mass Index (BMI) column.

```
{r}
# ii. Add BMI column and plot height against BMI
starwars <- starwars %>% mutate(BMI = mass / ((height / 100) ^ 2))

# Print the starwars
print(starwars)
```

A tibble: 87 × 15

height <int>	mass <dbl>	hair_color <chr>	skin_color <chr>	eye_color <chr>	birth_year <dbl>	sex <chr>	gender <chr>	homeworld <chr>	species <chr>	films <list>	vehicles <list>	starships <list>	BMI <dbl>
172	77.0	blond	fair	blue	19.0	male	masculine	Tatooine	Human	<chr [5]>	<chr [2]>	<chr [2]>	26.02758
167	75.0	NA	gold	yellow	112.0	none	masculine	Tatooine	Droid	<chr [6]>	<chr [0]>	<chr [0]>	26.89232
96	32.0	NA	white, blue	red	33.0	none	masculine	Naboo	Droid	<chr [7]>	<chr [0]>	<chr [0]>	34.72222
202	136.0	none	white	yellow	41.9	male	masculine	Tatooine	Human	<chr [4]>	<chr [0]>	<chr [1]>	33.33007
150	49.0	brown	light	brown	19.0	female	feminine	Alderaan	Human	<chr [5]>	<chr [1]>	<chr [0]>	21.77778
178	120.0	brown, grey	light	blue	52.0	male	masculine	Tatooine	Human	<chr [3]>	<chr [0]>	<chr [0]>	37.87401
165	75.0	brown	light	blue	47.0	female	feminine	Tatooine	Human	<chr [3]>	<chr [0]>	<chr [0]>	27.54821
97	32.0	NA	white, red	red	NA	none	masculine	Tatooine	Droid	<chr [1]>	<chr [0]>	<chr [0]>	34.00999
183	84.0	black	light	brown	24.0	male	masculine	Tatooine	Human	<chr [1]>	<chr [0]>	<chr [1]>	25.08286
182	77.0	auburn, white	fair	blue-gray	57.0	male	masculine	Stewjon	Human	<chr [6]>	<chr [1]>	<chr [5]>	23.24598
188	84.0	blond	fair	blue	41.9	male	masculine	Tatooine	Human	<chr [3]>	<chr [2]>	<chr [3]>	23.76641
180	NA	auburn, grey	fair	blue	64.0	male	masculine	Eriadu	Human	<chr [2]>	<chr [0]>	<chr [0]>	NA
228	112.0	brown	unknown	blue	200.0	male	masculine	Kashyyyk	Wookiee	<chr [5]>	<chr [1]>	<chr [2]>	21.54509
180	80.0	brown	fair	brown	29.0	male	masculine	Corellia	Human	<chr [4]>	<chr [0]>	<chr [2]>	24.69136
173	74.0	NA	green	black	44.0	male	masculine	Rodia	Rodian	<chr [1]>	<chr [0]>	<chr [0]>	24.72518
175	1358.0	NA	green-tan, brown	orange	600.0	hermaphroditic	masculine	Nal Hutta	Hutt	<chr [3]>	<chr [0]>	<chr [0]>	443.42857
170	77.0	brown	fair	hazel	21.0	male	masculine	Corellia	Human	<chr [3]>	<chr [1]>	<chr [1]>	26.64360
180	110.0	brown	fair	blue	NA	NA	NA	Bestine IV	NA	<chr [1]>	<chr [0]>	<chr [1]>	33.95062
66	17.0	white	green	brown	896.0	male	masculine	NA	Yoda's species	<chr [5]>	<chr [0]>	<chr [0]>	39.02663
170	75.0	grey	pale	yellow	82.0	male	masculine	Naboo	Human	<chr [5]>	<chr [0]>	<chr [0]>	25.95156
183	78.2	black	fair	brown	31.5	male	masculine	Kamino	Human	<chr [3]>	<chr [0]>	<chr [1]>	23.35095
200	140.0	none	metal	red	15.0	none	masculine	NA	Droid	<chr [1]>	<chr [0]>	<chr [0]>	35.00000
190	113.0	none	green	red	53.0	male	masculine	Trandosha	Trandoshan	<chr [1]>	<chr [0]>	<chr [0]>	31.30194
177	79.0	black	dark	brown	31.0	male	masculine	Socorro	Human	<chr [2]>	<chr [0]>	<chr [1]>	25.21625
175	79.0	none	light	blue	37.0	male	masculine	Bespin	Human	<chr [1]>	<chr [0]>	<chr [0]>	25.79592
180	83.0	none	brown mottle	orange	41.0	male	masculine	Mon Cala	Mon Calamari	<chr [2]>	<chr [0]>	<chr [0]>	25.61728
150	NA	auburn	fair	blue	48.0	female	feminine	Chandrila	Human	<chr [1]>	<chr [0]>	<chr [0]>	NA
NA	NA	brown	fair	brown	NA	male	masculine	NA	Human	<chr [1]>	<chr [0]>	<chr [1]>	NA
88	20.0	brown	brown	brown	8.0	male	masculine	Endor	Ewok	<chr [1]>	<chr [0]>	<chr [0]>	25.82645
160	68.0	none	grey	black	NA	male	masculine	Sullust	Sullustan	<chr [1]>	<chr [0]>	<chr [1]>	26.56250
193	89.0	brown	fair	blue	92.0	male	masculine	NA	Human	<chr [1]>	<chr [1]>	<chr [0]>	23.89326
191	90.0	none	mottled green	red	NA	male	masculine	Cato Neimoidia	Neimodian	<chr [3]>	<chr [0]>	<chr [0]>	24.67038
170	NA	blond	fair	blue	91.0	male	masculine	Coruscant	Human	<chr [1]>	<chr [0]>	<chr [0]>	NA
185	45.0	brown	light	brown	46.0	female	feminine	Naboo	Human	<chr [3]>	<chr [0]>	<chr [3]>	13.14828
196	66.0	none	orange	orange	52.0	male	masculine	Naboo	Gungan	<chr [2]>	<chr [0]>	<chr [0]>	17.18034
224	82.0	none	grey	orange	NA	male	masculine	Naboo	Gungan	<chr [1]>	<chr [0]>	<chr [0]>	16.34247
206	NA	none	green	orange	NA	male	masculine	Naboo	Gungan	<chr [1]>	<chr [0]>	<chr [0]>	NA
183	NA	brown	fair	blue	NA	male	masculine	Naboo	Human	<chr [1]>	<chr [0]>	<chr [1]>	NA
137	NA	black	blue, grey	yellow	NA	male	masculine	Toydaria	Toydarian	<chr [2]>	<chr [0]>	<chr [0]>	NA
112	40.0	none	grey, red	orange	NA	male	masculine	Malastare	Dug	<chr [1]>	<chr [0]>	<chr [0]>	31.88776
183	NA	black	dark	brown	62.0	male	masculine	Naboo	Human	<chr [1]>	<chr [0]>	<chr [0]>	NA
163	NA	black	fair	brown	72.0	female	feminine	Tatooine	Human	<chr [2]>	<chr [0]>	<chr [0]>	NA
175	80.0	none	red	yellow	54.0	male	masculine	Dathomir	Zabrak	<chr [1]>	<chr [1]>	<chr [1]>	26.12245
180	NA	none	pale	pink	NA	male	masculine	Ryloth	Twi'lek	<chr [1]>	<chr [0]>	<chr [0]>	NA
178	55.0	none	blue	hazel	48.0	female	feminine	Ryloth	Twi'lek	<chr [3]>	<chr [0]>	<chr [0]>	17.35892

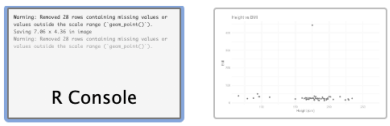
1–45 of 87 rows | 2–15 of 15 columns

Previous 1 2 Next

Plot of height versus BMI

```
# Plot height against BMI
ggplot(starwars, aes(x = height, y = BMI)) +
  geom_point() +
  labs(title = "Height vs BMI", x = "Height (cm)", y = "BMI") +
  theme_minimal()

# Save the plot
ggsave("height_vs_bmi.png")
```



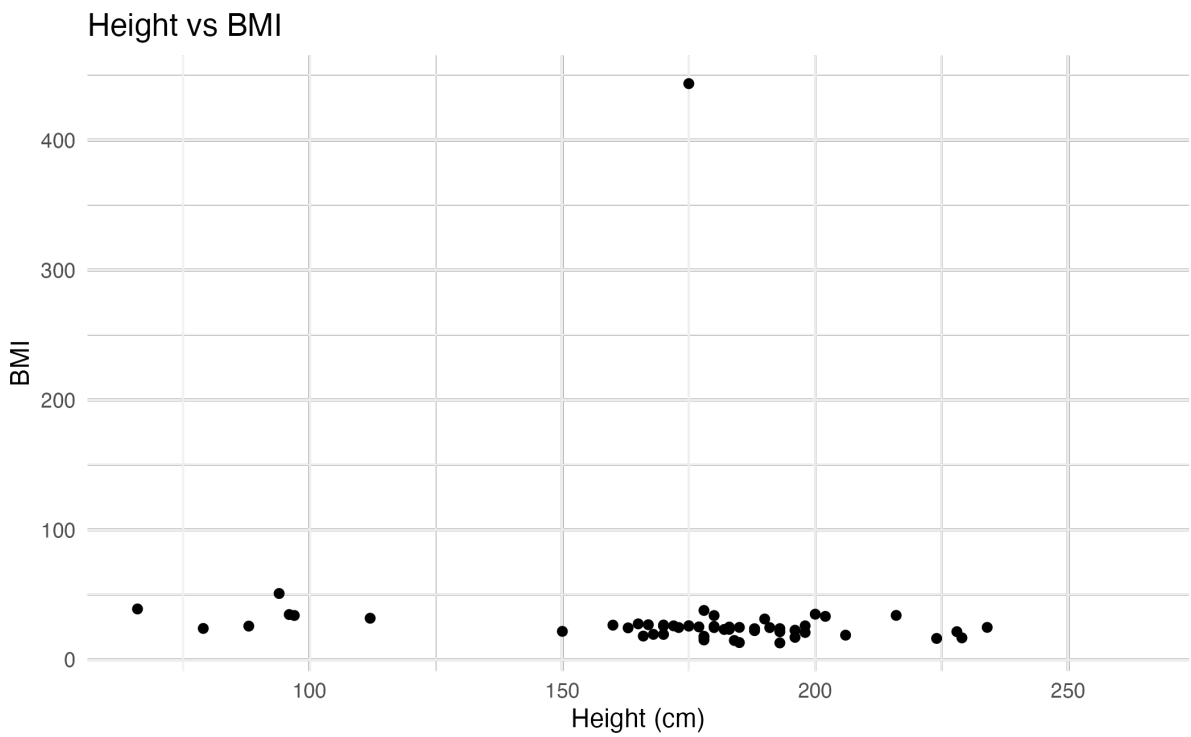

```
Warning: Removed 28 rows containing missing values or values outside the scale range (`geom_point()`).
Saving 7.06 x 4.36 in image
Warning: Removed 28 rows containing missing values or values outside the scale range (`geom_point()`).
```

```
{r}
# Identify rows with missing or out-of-range values
missing_values <- starwars %>% filter(is.na(height) | is.na(BMI))
print(missing_values)
```

A tibble: 28 × 15

name <chr>	height <int>	mass <dbl>	hair_color <chr>	skin_color <chr>	eye_color <chr>	birth_year <dbl>	sex <chr>	gender <chr>	homeworld <chr>	species <chr>	films <list>	vehicles <list>	starships <list>	BMI <dbl>
Wilhuff Tarkin	180	NA	auburn, grey	fair	blue	64	male	masculine	Eriadu	Human	<chr [2]>	<chr [0]>	<chr [0]>	NA
Mon Mothma	150	NA	auburn	fair	blue	48	female	feminine	Chandrila	Human	<chr [1]>	<chr [0]>	<chr [0]>	NA
Arvel Crynyd	NA	NA	brown	fair	brown	NA	male	masculine	NA	Human	<chr [1]>	<chr [0]>	<chr [1]>	NA
Finis Valorum	170	NA	blond	fair	blue	91	male	masculine	Coruscant	Human	<chr [1]>	<chr [0]>	<chr [0]>	NA
Rugor Nass	206	NA	none	green	orange	NA	male	masculine	Naboo	Gungan	<chr [1]>	<chr [0]>	<chr [0]>	NA
Ric Olié	183	NA	brown	fair	blue	NA	male	masculine	Naboo	Human	<chr [1]>	<chr [0]>	<chr [1]>	NA
Watto	137	NA	black	blue, grey	yellow	NA	male	masculine	Toydaria	Toydarian	<chr [2]>	<chr [0]>	<chr [0]>	NA
Quarsh Panaka	183	NA	black	dark	brown	62	male	masculine	Naboo	Human	<chr [1]>	<chr [0]>	<chr [0]>	NA
Shmi Skywalker	163	NA	black	fair	brown	72	female	feminine	Tatooine	Human	<chr [2]>	<chr [0]>	<chr [0]>	NA
Bib Fortuna	180	NA	none	pale	pink	NA	male	masculine	Ryloth	Twi'lek	<chr [1]>	<chr [0]>	<chr [0]>	NA
Gasgano	122	NA	none	white, blue	black	NA	male	masculine	Troiken	Xexto	<chr [1]>	<chr [0]>	<chr [0]>	NA
Eeth Koth	171	NA	black	brown	brown	NA	male	masculine	Iridonia	Zabrak	<chr [2]>	<chr [0]>	<chr [0]>	NA
Saesee Tiin	188	NA	none	pale	orange	NA	male	masculine	Iktotch	Iktotchi	<chr [2]>	<chr [0]>	<chr [0]>	NA
Yarael Poof	264	NA	none	white	yellow	NA	male	masculine	Quermia	Quermian	<chr [1]>	<chr [0]>	<chr [0]>	NA
Mas Amedda	196	NA	none	blue	blue	NA	male	masculine	Champala	Chagrian	<chr [2]>	<chr [0]>	<chr [0]>	NA
Cordé	157	NA	brown	light	brown	NA	NA	NA	Naboo	NA	<chr [1]>	<chr [0]>	<chr [0]>	NA
Cliegg Lars	183	NA	brown	fair	blue	82	male	masculine	Tatooine	Human	<chr [1]>	<chr [0]>	<chr [0]>	NA
Dormé	165	NA	brown	light	brown	NA	female	feminine	Naboo	Human	<chr [1]>	<chr [0]>	<chr [0]>	NA
Bail Prestor Organa	191	NA	black	tan	brown	67	male	masculine	Alderaan	Human	<chr [2]>	<chr [0]>	<chr [0]>	NA
Taun We	213	NA	none	grey	black	NA	female	feminine	Kamino	Kaminoan	<chr [1]>	<chr [0]>	<chr [0]>	NA
Jocasta Nu	167	NA	white	fair	blue	NA	female	feminine	Coruscant	Human	<chr [1]>	<chr [0]>	<chr [0]>	NA
R4-P17	96	NA	none	silver, red	red, blue	NA	none	feminine	NA	Droid	<chr [2]>	<chr [0]>	<chr [0]>	NA
San Hill	191	NA	none	grey	gold	NA	male	masculine	Muunilinst	Muun	<chr [1]>	<chr [0]>	<chr [0]>	NA
Finn	NA	NA	black	dark	dark	NA	male	masculine	NA	Human	<chr [1]>	<chr [0]>	<chr [0]>	NA
Rey	NA	NA	brown	light	hazel	NA	female	feminine	NA	Human	<chr [1]>	<chr [0]>	<chr [0]>	NA
Poe Dameron	NA	NA	brown	light	brown	NA	male	masculine	NA	Human	<chr [1]>	<chr [0]>	<chr [1]>	NA
BB8	NA	NA	none	none	black	NA	none	masculine	NA	Droid	<chr [1]>	<chr [0]>	<chr [0]>	NA
Captain Phasma	NA	NA	none	none	unknown	NA	female	feminine	NA	Human	<chr [1]>	<chr [0]>	<chr [0]>	NA

28 rows



The graph plots Height (in centimeters) on the x-axis against BMI (Body Mass Index) on the y-axis for characters from the Star Wars dataset. Here's a breakdown of what the graph shows:

1. **Overall Trend**: Most data points are clustered at the bottom of the graph, indicating that the BMI values for the majority of the characters are relatively low. The height of these characters ranges between approximately 150 cm and 220 cm.
2. **Outlier**: There is one significant outlier in the graph with a BMI much higher than the others. This character has a BMI of over 400, which is extremely high and unusual compared to the rest of the dataset.
3. **Height Distribution**: The height of the characters varies between around 65 cm and 250 cm, but most characters have heights between 150 cm and 200 cm.

4. **BMI Distribution**: Except for the outlier, the BMI values are mostly clustered below 100. This suggests that, for the majority of characters, their weight and height result in a lower BMI, which is consistent with normal to overweight categories in humans.

Possible Interpretation:

- The outlier could represent a character with an unusual physique or an error in the data entry, as such a high BMI is not typical.
- The rest of the characters follow a more expected distribution, where the BMI increases slightly with height but remains within a typical range.
- Characters that are shorter generally have a slightly lower BMI, which might indicate that they are lighter relative to their height.

The dataset seems to have a variety of characters with different body types, but the extreme outlier might require further investigation to understand if it's a valid data point or an anomaly.

```r
# Find the character with the highest BMI
highest_bmi <- starwars %>%
  filter(BMI == max(BMI, na.rm = TRUE))

# Display the character with the highest BMI
print(highest_bmi)
```

A tibble: 1 × 15

name <chr>	height <int>	mass <dbl>	hair_color <chr>	skin_color <chr>	eye_color <chr>	birth_year <dbl>	sex <chr>	gender <chr>	homeworld <chr>	species <chr>	films <list>	vehicles <list>	starships <list>	BMI <dbl>
Jabba Desilijic Tiure	175	1358	NA	green-tan, brown	orange	600	hermaphroditic	masculine	Nal Hutta	Hutt	<chr [3]>	<chr [0]>	<chr [0]>	443.4286

1 row

The outlier in the graph represents the character **Jabba Desilijic Tiure**, commonly known as Jabba the Hutt, from the Star Wars universe.

- **Height**: 175 cm
- **Mass**: 1358.0 kg
- **BMI**: 443.42857

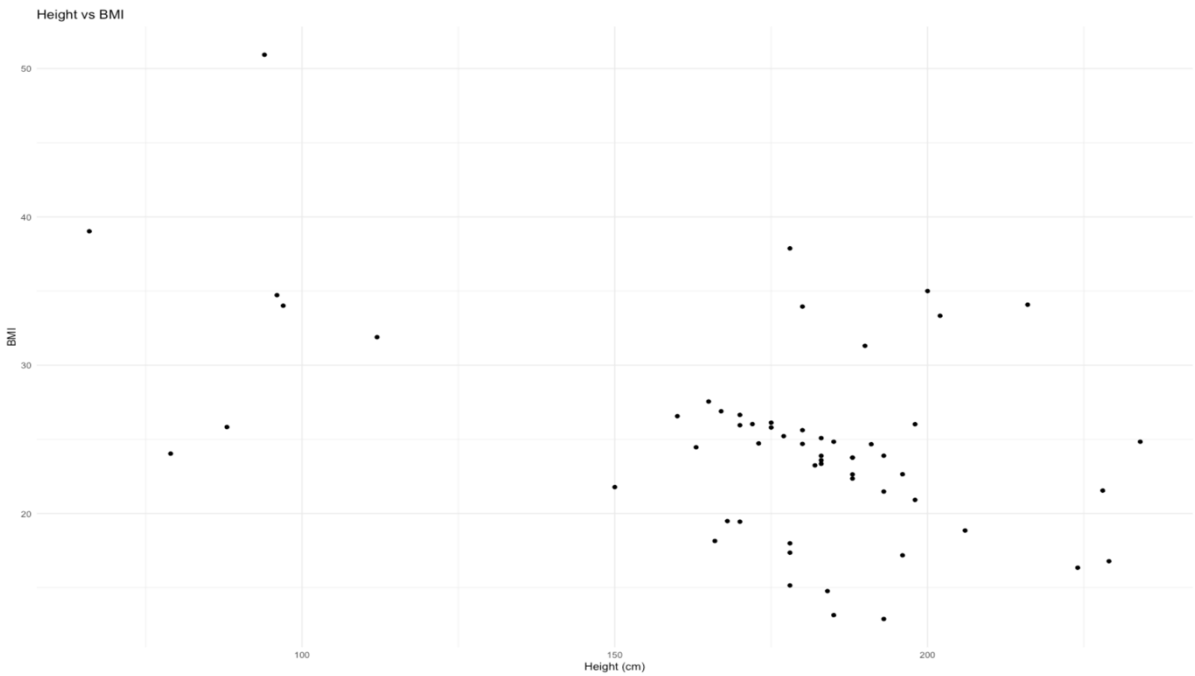
He has a body type that is extremely different from the more humanoid characters in the dataset. This leads to an exceptionally high BMI, which is reflected in the graph as an extreme outlier.

Jabba's data point appears far above the rest, showing that his BMI is an extreme anomaly. The majority of the characters have BMIs that are under 60), making Jabba's BMI of over 440 stand out significantly.

This also reflects how the BMI formula, designed for humans, might not accurately represent the body types of such non-humanoid creatures.

```r
# Find the character with the highest BMI
highest_bmi <- starwars %>%
  filter(BMI == max(BMI, na.rm = TRUE))

# Delete the row of the character with the highest BMI
starwars_without_outliner <- starwars %>%
  filter(BMI != max(BMI, na.rm = TRUE))

# Display the updated dataset
print(starwars_without_outliner)
```

```r
# Plot height against BMI of starwars_without_outliner
ggplot(starwars_without_outliner, aes(x = height, y = BMI)) +
  geom_point() +
  labs(title = "Height vs BMI", x = "Height (cm)", y = "BMI") +
  theme_minimal()

# Save the plot
ggsave("height_vs_bmi_without_outliner.png")
```